

A Basic VOLTAGE TO FREQUENCY CONVERTER

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A proven way to send an analogue signal a long distance is to transport the signal as a frequency using a voltage to frequency converter (VFC), a special circuit whose output is a frequency that is proportional to its input voltage.

In principle, it is relatively easy to send a frequency signal over a long transmission path without interference through optical, coaxial, or radio links. The frequency is then reconverted to an analogue voltage by a FVC that is usually tailored to perform its inverse function, often using a phase-locked loop (PLL). Block diagram of the project is shown in Fig. 1.

This project uses an odd approach to design and build a hobby-level VFC module using a handful

of easily available and inexpensive electronic components. Author's circuit of the VFC module wired on a breadboard is shown in Fig. 2.

Schematic diagram of the VFC module is shown in Fig. 3. Please note, a well-regulated 5V DC power source is absolutely necessary to run this VFC circuit. Heart of the circuit is the venerable CMOS micropower phase-locked loop IC HEF4046B.

The HEF4046B (IC1) includes a VCO sub-system in addition to phase comparators and other elements. It can generate a square wave with 50% duty cycle. The frequency is reasonably close to 0Hz-1kHz range, which is curbed by the 100nF capacitor (C1) and the 10k resistor (R1).

The output is available on pin 4

across CON3. The voltage input for the VCO on pin 9 (SEN) is connected across CON1. CON2 is used for connecting the 5V power supply.

As a quick aside, there's an op-

PARTS LIST

Semiconductors:

IC1 - HEF4046B phase-locked loop IC

LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 10-kilo-ohm

R2 - 1-kilo-ohm

VR1 - 100-kilo-ohm potmeter

Capacitors:

C1, C2 - 100nF disc

Miscellaneous:

CON1-CON3 - 2-pin connector

- Breadboard

Optional for Fig. 5 Circuit

Semiconductors:

IC2 - HEF4046B phase-locked loop IC

LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R3 - 10-kilo-ohm

R4 - 1-kilo-ohm

VR2 - 100-kilo-ohm potmeter

Capacitors:

C1, C2 - 100nF disc

Miscellaneous:

CON4-CON5 - 2-pin connector

- Breadboard

- LDR GL5528



Fig. 1: Block diagram of voltage to frequency converter

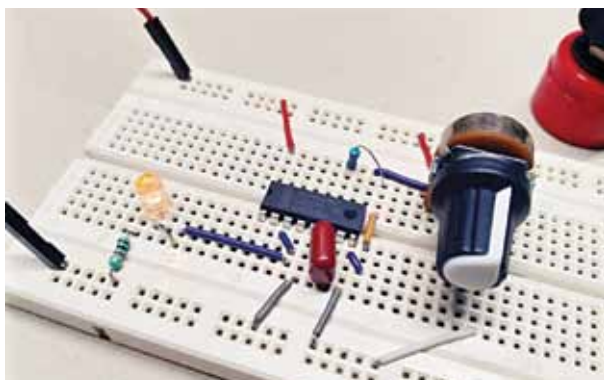


Fig. 2: Author's circuit on a breadboard

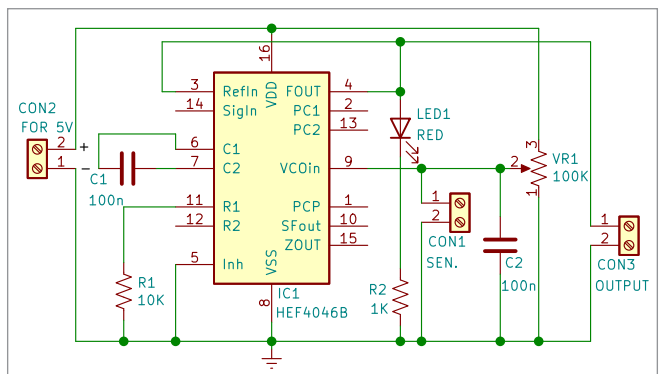


Fig. 3: Circuit diagram of the VFC module